

Research Gap Analysis: AI-Enabled Diagnosis of Pneumonia and Tuberculosis

Scope

This analysis reviews current literature across six technology areas — Digital Twins, Neurosymbolic/Explainable AI, IoT, Quantum AI, U-Net-based segmentation, and benchmark datasets — as applied to pneumonia and tuberculosis (TB), and synthesizes the resulting research gap(s).

1. Digital Twin (DT)

Current state. DT work for the lungs is real but thin and mostly recent (2022–2026). The most cited example is **Lung-DT**, which pairs a YOLOv8 classifier with IoT-sourced SpO₂ data to build a "digital representation" of a patient's respiratory health, classifying chest X-rays into normal/COVID/lung-opacity/pneumonia/TB with ~96.8% accuracy. Other efforts include a mechanistic DT of *Mycobacterium tuberculosis* infection (immune dynamics + pathogen adaptation, aimed at precision therapy) and ICU-focused DTs for mechanical-ventilation/pneumonia triage.

Gaps identified.

- A scoping review in the *European Respiratory Journal* found patient DTs for respiratory disease remain **few overall**, with only a handful covering pneumonia and essentially none built for TB at the patient (rather than pathogen) level.
 - Existing "digital twins" in the pneumonia/TB space are largely **static classification pipelines relabelled as DTs** — they classify a single X-ray plus a vital sign, rather than maintaining a continuously updated, bidirectional, patient-specific virtual model that evolves with new sensor/imaging data over time (the defining property of a true DT).
 - No DT couples the **infection-dynamics (mechanistic) TB models** with the **imaging/IoT-based respiratory DTs** — the immunological/pathogen-adaptation twins and the imaging-based twins are two disconnected literatures.
 - Reported DT datasets are small (tens to low thousands of patients) and mostly **"available upon request"**, not public — reproducibility and external validation are largely absent.
 - **Standardization gap:** healthcare-specific DT communication/data standards (e.g., ETSI SmartBAN) remain under-adopted, and there is no agreed architecture for how imaging, IoT vitals, and mechanistic infection models should be fused into one twin.
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2. Explainable AI — specifically Neurosymbolic AI

Current state. The overwhelming majority of "explainable AI" work on pneumonia/TB chest X-rays uses **post-hoc, purely statistical explanation methods** — Grad-CAM, LIME, SHAP — layered on top of CNNs (ResNet, DenseNet, VGG, EfficientNet) to produce heatmaps ("the model looked here"). Genuine **neurosymbolic AI** (combining neural

perception with symbolic/logical reasoning over medical knowledge, ontologies, or knowledge graphs) is an active general research area — surveys exist on neurosymbolic reasoning over biomedical knowledge graphs and on neurosymbolic AI for healthcare broadly — but almost none of this has been applied specifically to pneumonia or TB imaging.

Gaps identified.

- **No dedicated neurosymbolic pneumonia/TB system** was found: nothing that grounds CNN/ViT lung-image features in a formal clinical rule base (e.g., radiological criteria for consolidation, cavitation, Ghon foci) or a TB/pneumonia diagnostic ontology to produce a logically-derived, human-auditable diagnosis rather than a saliency map.
 - Current XAI outputs (heatmaps) show *where* a model looked but not *why* that finding implies pneumonia vs. TB vs. another pathology — they don't encode differential-diagnosis logic, so two diseases with overlapping opacity patterns can't be reliably distinguished through the explanation itself.
 - No integration of **radiology reporting standards/clinical guidelines as symbolic constraints** during training (neuro-symbolic loss functions, logic tensor networks) for lung disease — this exists conceptually in the broader neurosymbolic-healthcare literature but is untested on pneumonia/TB.
 - Explanation outputs are rarely **clinically validated with radiologists or TB clinicians** for trustworthiness/actionability, and there is no shared benchmark for "explanation quality" in this domain (unlike accuracy, which has many benchmarks).
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3. IoT

Current state. IoT applications split into two families: (a) **vital-sign/wearable monitoring** — SpO2, temperature, heart rate, respiratory rate sensors streamed via Wi-Fi/MQTT to a cloud dashboard for remote pneumonia patients (e.g., ICU/home monitoring systems, the ChARM device for pediatric pneumonia respiratory-rate counting in Mali); and (b) **breath/biomarker sensing** — e-nose/VOC sensors detecting TB-specific volatile biomarkers (methyl nicotinate, methyl anisate) transmitted via IoT for low-cost screening.

Gaps identified.

- IoT vital-sign systems and IoT breath-biomarker systems are developed as **separate, single-modality pipelines**; none combine physiological streams with breath-biomarker sensing and imaging in one continuous pneumonia/TB monitoring loop.
- Most IoT-TB breath-sensor studies are **small proof-of-concept lab prototypes**, not clinically validated against sputum smear/GeneXpert/culture at scale, and lack large public datasets of paired sensor readings and confirmed diagnoses.
- Very little work addresses **low-resource-setting deployment constraints** (intermittent connectivity, power, cost, data governance) even though pneumonia and TB burden is concentrated in low- and middle-income countries — most IoT prototypes assume continuous Wi-Fi/cloud access.

- No standardized **security/privacy framework** for continuous physiological + imaging IoT data streams specific to infectious respiratory disease monitoring (contrasted with the generic IoT security literature).
 - Weak linkage between IoT monitoring output and downstream **AI diagnostic/DT models** — the sensor data and the imaging-classification models are usually evaluated in isolation rather than as one fused decision pipeline (Lung-DT is a rare exception, and even there fusion is shallow — SpO2 is an auxiliary input, not deeply integrated).
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4. Quantum AI

Current state. Work is early-stage and sparse. Notable examples: **quanvolutional neural networks (QNNs)** — hybrid quantum-classical CNNs — applied to TB screening on chest X-rays (meta-analysis, 2025); a **Quantum Boltzmann Machine (QBM)** applied to pediatric pneumonia detection on the MedMNIST dataset (QuCUN platform, LMU/Aqarios/BASF/SAP), reporting ~84-86% accuracy with under 9,000 trainable parameters; and general quantum-SVM/quantum hybrid architectures explored for related lung-disease image classification (e.g., lung cancer).

Gaps identified.

- Quantum AI for pneumonia/TB is confined almost entirely to **classification benchmarking on small/simulated quantum hardware or classical simulators** — no work addresses segmentation, localization, differential diagnosis, or clinical deployment.
 - Reported accuracies (mid-80s%) still **trail classical CNN/transformer baselines** (which reach mid-90s to high-90s%), so the literature has not yet demonstrated where quantum advantage (fewer parameters, faster training, better small-data generalization) translates into superior *clinical* performance rather than just parameter efficiency.
 - No study benchmarks quantum models on the **standard TB/pneumonia public datasets** (Shenzhen, Montgomery, RSNA) used by the classical literature — most use small/synthetic sets (e.g., MedMNIST), making cross-comparison with classical SOTA impossible.
 - **No explainability layer exists for quantum or hybrid quantum-classical models** in this domain — quantum AI and XAI/neurosymbolic AI are currently disconnected research threads.
 - Essentially nothing addresses **TB specifically at scale** with quantum methods beyond the single quanvolutional-CNN meta-analysis; pneumonia quantum work is similarly limited to one or two proof-of-concept studies.
 - Real quantum-hardware validation (vs. simulators) is absent, and there is no cost/latency/feasibility analysis for real-world (e.g., point-of-care) deployment.
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5. Segmentation Algorithm — U-Net

Current state. U-Net and its variants (Attention U-Net, U-Net++, ResU-Net, nnU-Net, RINGU-NET, DEFU-Net) are the dominant architecture for lung-field and lesion segmentation in both CXR and CT, used as a preprocessing step before classification for both pneumonia and TB. Variants report high Dice scores on standard TB datasets (e.g., RINGU-NET: Dice 97.2% on Shenzhen).

Gaps identified.

- Plain U-Net's **skip connections cause a semantic gap between encoder and decoder** and lose spatial detail through repeated downsampling, leading to inconsistent performance on X-rays with **complex pathological opacities** — precisely the abnormal cases (dense consolidation, cavitation, miliary patterns) that matter most for pneumonia/TB — as opposed to normal lungs, where segmentation is comparatively easy.
- A dedicated **TB-consistent lesion (ROI) segmentation** (as opposed to whole-lung-field segmentation) remains largely unsolved: as of the cited 2021 study, "there are no studies... that perform CXR-based TB-consistent ROI segmentation," and this remains a thin literature relative to whole-lung segmentation.
- Performance depends heavily on **manually annotated pixel-level masks**, which are scarce and expensive in medical imaging — most TB/pneumonia segmentation datasets provide only lung-field masks (Montgomery/Shenzhen), not lesion-level masks, forcing weak/self-supervision workarounds.
- **Cross-manufacturer / cross-population generalization** is weak: models trained on one dataset (e.g., Shenzhen) degrade on another (e.g., Montgomery) due to scanner/resolution/demographic differences — only a handful of studies (e.g., DEFU-Net) explicitly test cross-dataset segmentation.
- Segmentation and downstream classification/DT/XAI stages are usually **evaluated as separate, disconnected steps** rather than jointly optimized or evaluated for how segmentation errors propagate into diagnostic or explanation errors.
- 3D CT lesion segmentation (as opposed to 2D CXR) for TB/pneumonia is comparatively under-studied relative to CXR, despite CT's superior sensitivity for early/atypical disease.

6. Benchmark Datasets

The most commonly used public benchmark datasets for pneumonia and TB imaging AI research are:

Dataset	Disease	Modality	Size	Annotation	Notes
Shenzhen & Montgomery (NLM/MC-CHN) TB sets	TB	CXR	662 (Shenzhen) + 138 (Montgomery) images	Image-level labels; lung-field segmentation masks	Most widely used TB benchmark pair; small, low patient diversity
RSNA Pneumonia Detection Challenge dataset	Pneumonia	CXR (DICOM)	~26,000–30,000 images	Bounding boxes for lung opacities	Derived from NIH ChestX-ray14; large but pneumonia-only, no TB
NIH ChestX-ray14 / CheXpert / PadChest / VinDr-CXR / MIMIC-CXR	Pneumonia (as 1 of 14+ labels)	CXR	100,000+ images	Weak, NLP-extracted labels from reports	Large-scale but noisy labels; pneumonia is a minority class among many findings
Kaggle "Chest X-Ray Images (Pneumonia)" (Kermany et al.)	Pediatric pneumonia	CXR	~5,863 images	Image-level (bacterial/viral/normal)	Small, single pediatric hospital, known selection-bias issues
Belarus TB Portal / NIAID TB Portal	TB	CXR	~300–3,000 images	Image-level; some drug-resistance metadata	Access partially restricted; useful for cross-dataset generalization tests

Gaps identified.

- No large-scale, **public, pixel-annotated dataset combining pneumonia and TB** with lesion-level (not just lung-field) masks exists — researchers must currently stitch together disparate small datasets (as Lung-DT and others do).

- TB datasets (Shenzhen/Montgomery/Belarus) are **an order of magnitude smaller** than pneumonia/general CXR datasets (RSNA, CheXpert, MIMIC-CXR), creating a data-imbalance problem that limits deep model generalization for TB specifically.
 - Almost all datasets are **2D CXR only**; public, annotated CT datasets for pneumonia/TB lesion segmentation are scarce (e.g., ImageCLEF Tuberculosis is one of the few, and is limited in size/access).
 - Very few datasets pair imaging with **IoT/physiological sensor data or longitudinal follow-up**, which is what a digital twin or IoT-driven monitoring system would require — imaging datasets and vitals-monitoring datasets exist in separate silos.
 - **Demographic and geographic diversity is limited** — many datasets originate from a handful of sites (Shenzhen, Montgomery County, a few Kaggle sources), raising external-validity concerns for global deployment, especially in high-TB-burden, low-resource countries.
 - No dataset currently includes **clinician-authored explanations, rationale annotations, or symbolic/ontology-linked labels**, which would be needed to train or evaluate neurosymbolic XAI approaches.
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Synthesized Cross-Cutting Research Gap

Taken individually, each technology area shows measurable but incremental progress on pneumonia/TB. The **larger, more consequential gap is integration**: every reviewed study addresses at most one or two of these six dimensions in isolation, and no work brings them together into a single validated pipeline. Specifically:

1. **No end-to-end system exists** that couples (a) precise **lesion-level segmentation** (going beyond plain U-Net's limitations on abnormal/complex lung findings), with (b) a **neurosymbolic explanation layer** grounded in clinical/ontological knowledge rather than post-hoc saliency maps, feeding into (c) a **continuously updated digital twin** of the patient that ingests (d) **real-time IoT physiological/breath-biomarker data**, while exploring whether (e) **quantum-enhanced models** offer genuine advantage (not just parameter efficiency) — all trained and validated on (f) a **large-scale, diverse, pixel- and biomarker-annotated public benchmark** spanning both pneumonia and TB.
2. **TB is consistently the underserved disease** relative to pneumonia/COVID-19 across all six areas — smaller datasets, fewer DTs, fewer dedicated segmentation models, and almost no quantum-AI work — despite TB's disproportionate burden in low-resource settings where IoT/low-cost screening would matter most.
3. **Low-resource-setting deployability** (connectivity, cost, hardware, annotation burden) is treated as an afterthought rather than a design constraint in nearly all of the reviewed literature, even though pneumonia and TB mortality is concentrated precisely in such settings.
4. **Explainability and trust remain unresolved at the clinical level**: existing XAI is visual/statistical, not reasoning-based, and has rarely been validated with

radiologists/TB clinicians for real diagnostic utility — a genuine neurosymbolic approach validated on pneumonia/TB has not yet been demonstrated.

Proposed research gap statement: *There is no validated, integrated framework that combines robust lesion-level segmentation, knowledge-grounded (neurosymbolic) explainability, real-time IoT-based physiological/biomarker monitoring, and digital-twin-style continuous patient modeling for pneumonia and tuberculosis — particularly one benchmarked on a large, diverse, richly annotated public dataset and designed for deployment in low-resource, high-burden settings, with quantum-AI components evaluated rigorously against classical baselines rather than in isolation.*

Compiled from a targeted review of peer-reviewed and preprint literature (2020–2026) via web search; individual claims are traceable to the cited studies discussed above.